TRACEABILITY AND DATA SOVEREIGNTY

SFSCON 2026

WHO OWNS YOUR AI DATA PIPELINE?

Open standards, lineage, and lakehouse architectures for modern AI systems

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HI, I'M SAMANTHA CRUZ

- Data Engineering Team Lead at isolutions
- Naval engineer by training, from oceanography research: satellite images, ocean models, then data
- CI/CD, data architecture, data engineering and data analysis
- Builds open-source, traceable AI pipelines on the side



BASED IN BARCELONA

EVERYONE WANTS TO CHAT WITH DATA

The most requested AI data pipeline today

- **Easy to build:** a vector store, an embedding model and an API key
- **Easy to demo:** it works the same afternoon
- **Everyone asks:** managers, customers and colleagues
- **Hard to explain:** almost nobody can say where the data goes

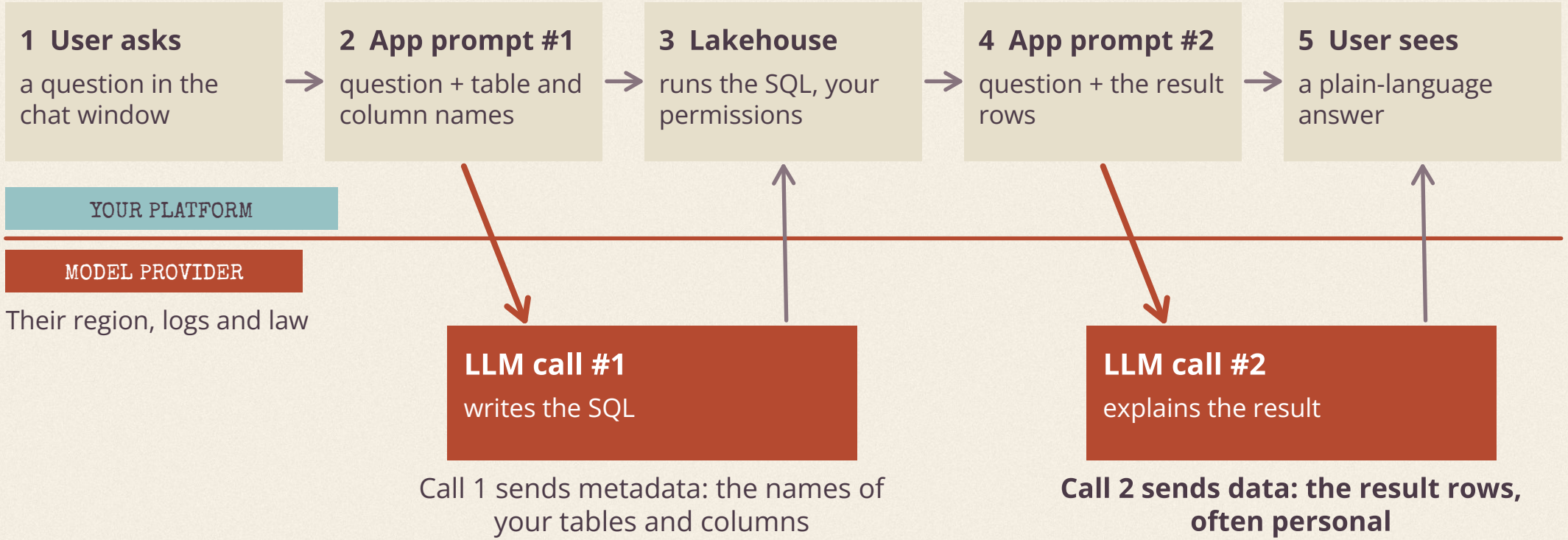


THE CATCH

The demo hides the data flow.

YOUR DATA GOES TO THE MODEL TWICE

Text-to-SQL chatbot, step by step



TAKEAWAY **Call 2 is the one teams forget: it sends the data itself.**

THREE QUESTIONS MOST TEAMS CAN'T ANSWER

A show of hands

1

WHICH COUNTRY?

- Where was my last prompt processed?

2

HOW LONG KEPT?

- How long does the provider keep it?

3

CAN YOU PROVE IT?

- Could you show an auditor tomorrow?

OWNING DATA MEANS THREE ABILITIES



READ IT

- Open formats: Parquet, Delta, Iceberg
- Any engine can open it



MOVE IT

- Names, grants, history exportable
- A catalog you control



EXPLAIN IT

- Trace answers to sources
- Lineage events you store

TAKEAWAY **Prompts, chunks and answers are data too.**

YOUR PLAN DECIDES, NOT THE MODEL

Checked November 2026

HOW YOU USE IT	USED FOR TRAINING?	KEPT FOR
Consumer app	Your toggle decides	30 days; 5 years if opted in
Business plan	No, by default	Set by contract and admin
API, direct	No, by default	Short; zero retention on request
API via cloud	No	Bedrock says not stored

TAKEAWAY **A consumer toggle is a promise, not a contract.**

TRAINING IS ONLY HALF THE RISK

- **Safety flags:** flagged chats may be kept up to 2 years, even with zero retention
- **Feedback:** in Claude's consumer app, a thumbs-up keeps the whole chat up to 5 years
- **Courts:** a 2025 US order made OpenAI keep logs; zero-retention API was exempt
- **Free tools:** staff pasting work data into consumer apps bypasses every control



WHY IT MATTERS

Zero retention leaves nothing to preserve.

AN EU REGION ANSWERS ONE QUESTION



RESIDENCY

- Where is it processed and stored?
- Answered by region settings



RETENTION

- How long, and used for training?
- Answered by plan and contract



JURISDICTION

- Whose law can compel access?
- Answered by who owns the provider

TAKEAWAY **A region setting cannot change who owns the provider.**

CLAUDE AND GPT, KEPT IN EUROPE

Where it runs, and what you can name

ROUTE	RUNS IN THE EU?	WATCH OUT FOR
Anthropic API, direct	No EU option	Only US or global
Claude on Bedrock	Yes, EU profiles	Pin a region for one country
Claude on Vertex AI	Yes, EU multi-region	EU member states only
OpenAI API, Europe	Yes, Europe project	Region fixed at creation
Azure OpenAI, Data Zone	Yes, anywhere in the EU	Not your resource's region
Mistral API	Yes, Sweden per help centre	EU-owned company

TAKEAWAY **Usually yes for the EU, but a zone, not a country.**

THREE NUMBERS TO BUDGET

+10%

Per token for residency

OpenAI, Azure, Vertex, Mistral

$0.02

Per GB between regions

AWS US list price

~$2k

Per GPU, per month

Rented H100, running 24/7

Sources: Digital Applied (Aug 2026); AWS Architecture Blog (US prices, 2021); Lyceum (Oct 2026, $2.50 to $3.20 per GPU-hour). Confirm current prices.

TRUST, THEN VERIFY

Four layers of evidence

01 

CONTRACT

Terms, data processing agreement, zero retention amendment

02 

CONFIG

Pin regions, block global routes, restrict outbound traffic

03 

RUNTIME

Read the region the API reports and the cloud audit logs

04 

LINEAGE

Record every model call in events you store yourself

TAKEAWAY **You can verify residency and retention. Jurisdiction needs the ladder.**

LINEAGE TURNS PROMISES INTO EVIDENCE



LOG EVERY CALL

- Provider, model, version
- Region and retention mode
- Embedding calls included



STORE HASHES

- Prompt hash, chunk IDs
- Never raw personal data



OPEN STANDARD

- OpenLineage events
- Kept in tables you own

TAKEAWAY **If you can't query it, you can't audit it.**

SEAL: THE WEAKEST LINK SETS THE LEVEL

The Commission's Cloud Sovereignty Framework

- **Eight objectives:** legal, data and AI, operations, supply chain, technology and more
- **Lowest wins:** your overall level is the lowest you reach on any objective
- **Grades offers:** it scores cloud services in a tender, not companies
- **First results:** Scaleway, STACKIT and Post Telecom's group reached SEAL-3



THE LADDER FOR AN AI PIPELINE

SEAL-4	**Full control**	EU-owned stack, open models on your own GPUs
SEAL-3	**Portable**	EU-operated cloud, swappable open-weight models
SEAL-2	**Pinned in the EU**	EU endpoints, zero retention, your keys and lineage
SEAL-1	**Under contract**	Business plan, no training, but global routing
SEAL-0	**Unmanaged**	Consumer apps, retention unknown



NOTE

My engineering reading, not the Commission's text.

EACH STEP HAS A TECHNICAL PRICE

STEP	HARDEST PART	COST SIGNAL
0 to 1	Finding shadow AI use	Seats and policy work
1 to 2	Every feature must honour the region	About 10% on tokens, plus migration
2 to 3	Portable stack, re-tested prompts, GPUs	About $2k a month per GPU
3 to 4	Chips and firmware	Not on sale today

TAKEAWAY **The cheapest step that matters most is 1 to 2.**

MOVING REGIONS IS A MIGRATION

Example: from a global setup to an EU region

01 

NEW PROJECT

The region is fixed at creation, so create a new one

02 

RE-EMBED

Rebuild vectors and indexes if the embedding model changes

03 

RUN BOTH

Duplicate storage, and transfer between regions is charged per GB

04 

CUT OVER

Switch traffic, verify in the logs, delete the old copy

TAKEAWAY **EU Data Act bans switching fees from 12 January 2027.**

THE MINIMUM TO ASK OF ANY PIPELINE

- **Classify:** which data may leave your platform, and which may leave the EU
- **Route:** business plan or API for work data, never a consumer app
- **Pin:** regions and endpoints in configuration, embedding call included
- **Trace:** zero retention where possible, every call logged as lineage
- **Ask vendors:** where weights run, who reads logs, how long, trained on?



EXIT TEST

If you can't switch the model, you don't own the hop.

OPEN USE CASES YOU CAN RUN

Built around the ideas in this talk

- **jobradar:** local models, lineage on every step, data stays on your machine
- **A second use case:** for sales, built the same way
- **Fully open source:** everything is on my GitHub



sammcruz.github.io

THANK YOU